Md. Jubair Uddin Zidan

Mechanical Engineering Researcher • Computational Fluid Dynamics & Architected Metamaterials

Adabor, Shyamoli, Dhaka, Bangladesh | Email: jubairzidan99@gmail.com | Institution: CUET
LinkedIn: linkedin.com/in/jubairzidan | Scholar: Md. Jubair Uddin Zidan | GitHub: github.com/jubairzidan

EDUCATION

Master of Science (M.Sc.) in Mechanical Engineering
Chittagong University of Engineering and Technology (CUET)
2025 – 2026
Ongoing Postgraduate

- Theory Coursework Cumulative CGPA: **3.90 / 4.00**
- Focus: Advanced Computational Fluid Dynamics, Turbulence Modeling, and Metamaterials

Bachelor of Science (B.Sc.) in Mechanical Engineering
Chittagong University of Engineering and Technology (CUET)
2019 – 2024
Degree Conferred

- Cumulative GPA across Upper Semesters: **3.41 / 4.00** (Graduating Class: 2024)
- Senior Capstone Thesis: High-Fidelity 3D Ground Effect Aerodynamics of Modern F1 Venturi Tunnels

RESEARCH EXPERIENCE & PUBLICATIONS

Undergraduate Bachelor Thesis: Aerodynamics & Ground Effect CFD
Chittagong University of Engineering and Technology (CUET), Dept. of Mechanical Engineering | 2024

- Conducted full-domain 3D computational fluid dynamics (CFD) simulations investigating Venturi ground-effect floors under modern FIA Formula One technical regulations.
- Analyzed vortex breakdown phenomena, suction peak preservation, downforce generation, and sensitivity to ride-height variations.

Current Manuscript: Architected Auxetic Mechanical Metamaterials for Energy Absorption
Under Review | Authors: Jubair Zidan, Wakil Ahmed, Dibosh Das | Supervisor: Dr. Kazi Zahir Uddin (UIUC / CUET)

- Systematic state-of-the-art review and numerical characterization of negative Poisson's ratio cellular metamaterials (2D re-entrant, chiral, and 3D rotational lattices).
- Evaluated specific energy absorption (SEA), dynamic impact crashworthiness, and multi-objective topology optimization.

Conference Paper: Aerodynamic Characterization of Bionic Tubercles on Airfoils
International Conference on Mechanical Engineering and Renewable Energy (ICMERE)

- Investigated stall delay mechanisms and boundary layer vortex generation across modified NACA airfoils under varying Reynolds numbers.

TECHNICAL & COMPUTATIONAL SKILLS

CFD & FEA Toolkits:	ANSYS Fluent, ANSYS Workbench, OpenFOAM, COMSOL Multiphysics, Abaqus
CAD & Design:	SolidWorks, Autodesk Fusion 360, AutoCAD, SpaceClaim
Programming & Math:	MATLAB, Python (NumPy, SciPy, Matplotlib), C++, LaTeX
Specialized Domains:	Aerodynamics, Aeroacoustics, Turbulence Modeling, Metamaterials, Topology Optimization

PROFESSIONAL & LEADERSHIP EXPERIENCE

ASME Student Section CUET — Executive Committee Member (2022 – 2024)

- Organized technical symposia, CAD modeling challenges, and national student engineering conventions.

Industrial Engineering Internship — Thermal Power Generation Plant (2023)

- Hands-on attachment analyzing combined cycle gas turbine operations, heat recovery steam generators (HRSG), and maintenance protocols.

Academic Instructorship & STEM Mentor — Physics & Higher Mathematics (2021 – Present)

- Taught Newtonian mechanics, electromagnetism, and advanced calculus to aspiring undergraduate engineering candidates.

HONORS, AWARDS & RECOGNITION

- **Champion — National Mechanical Engineering Innovation Design Challenge**
- **Scholarly Poster Symposium Award** for CFD Ground-Effect Investigations
- **Global Student Leadership Summit Delegate** representing CUET Faculty of Mechanical Engineering